# Flood Risk Early-Warning Model

for IoT-Based Pond Management System

Technical Report — Data Collection, Model Development, and Evaluation


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This report summarizes the data collection, feature engineering, and modeling work completed for the flood-risk component of our IoT pond management capstone project. It covers two independently trained models built from the same underlying dataset, and closes with the limitations and open questions we would like our advisor's guidance on.

# 1. Introduction

Our final-year project is an IoT-based pond management system for aquaculture, combining pH, turbidity, temperature, ultrasonic water-level, and soil sensors with two motors that exchange pond water when conditions become unsafe. That system is **reactive** — it senses and responds to conditions that already exist at the pond.

To extend the system with a **proactive** layer, we built a machine learning model that estimates regional flood risk 24 to 72 hours in advance, using free, publicly available hydrological and satellite data for Bangladesh. The goal is to give a fish/pond farmer real lead time to prepare — before the pond's own sensors detect a problem, not only after.

This report documents how we collected and validated our training data, how we built and evaluated two different models on that data, and the real limitations and open decisions we would like advice on going forward.

# 2. Data Collection

## 2.1 Monitoring Network

We defined 30 virtual gauge stations across Bangladesh's four major river systems (Brahmaputra/Jamuna, Ganges/Padma, Meghna/Surma-Kushiyara, and the Chittagong Hill Tracts), chosen to match real FFWC (Flood Forecasting & Warning Centre) monitoring points and cover every major basin rather than concentrating on one river.

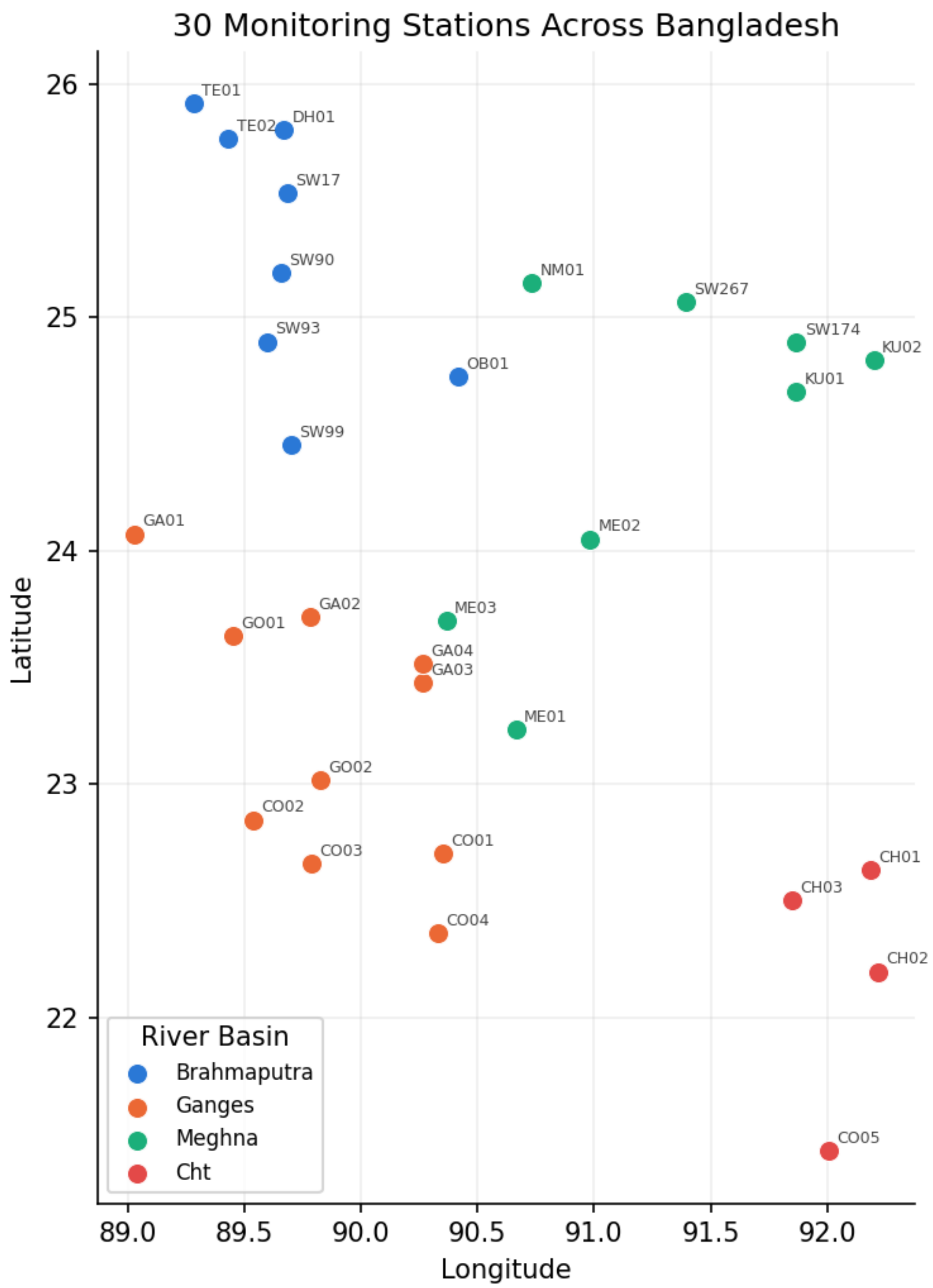


*Figure 1. The 30 monitoring stations used in this project, colored by river basin.*

## 2.2 Data Sources

We combined data from multiple free, public sources rather than a single provider, since no single free source covers both long history and reliable flood labels:

| Data type | Source | Coverage | Role |
| --- | --- | --- | --- |
| Rainfall & soil moisture | Open-Meteo (ERA5/ERA5-Land reanalysis) | 1950 – present | Feature |
| River discharge | Open-Meteo Flood API (GloFAS v4 reanalysis) | ~1997 – present | Feature |
| Terrain (elevation, HAND) | MERIT Hydro (via Google Earth Engine) | Static | Feature |
| Flood detection (primary) | NASA GFMS satellite flood intensity | 2013–2016, 2021–2026 | Label |

| | | | |
|---|---|---|---|
| Historical flood events | Dartmouth Flood Observatory (DFO) | 1985 – 2010 | Label (positive-only) |
| Historical flood events | Global Flood Database (MODIS) | 2003 – 2017 | Label (positive-only) |
| SAR flood extent | Copernicus Global Flood Monitoring (Sentinel-1) | 2016 – 2020 | Label (positive-only) |
| Official flood reports | FFWC Annual Flood Reports (parsed from PDF) | 2012 – 2021 | Label (positive-only) |

**Why multiple label sources:** NASA's GFMS satellite product is our most reliable label source, but its archive is only fully accessible for two windows (2013–2016 and 2021–2026); outside those windows we cannot confidently say a day was flood-free. The four additional sources extend our labeled history back to 1985, but each of them can only confirm a flood *did* happen — none of them catalogue every non-flood day, so they contribute positive examples only, never negative ones. We treat these two label types differently during training (Section 3.3).

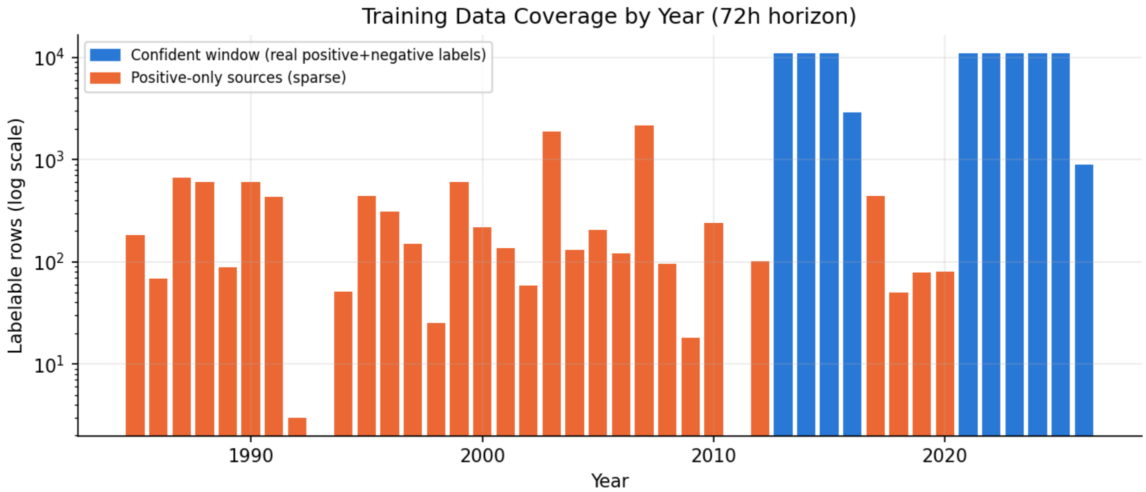


*Figure 2. Labelable training rows per year (log scale). Blue years have both real positive and negative labels (dense, reliable); orange years come only from positive-only sources (sparse — a match is trusted, but an unlabeled day is not evidence the area stayed dry).*

In total we assembled 839,340 station-days of raw feature data, of which 101,704 have a usable flood label for our longest (72-hour) prediction horizon, and 324,322 have a usable river-discharge value for our regression model (Section 4).

## 2.3 Sample Data: What "Good," "Incomplete," and "Bad" Look Like in Practice

Real hydrological data is never uniformly clean. Rather than only describing our data quality process in the abstract, this section shows three real examples from our own pipeline.

### (a) Good — a complete, reliable training row

A row from our densest, most trustworthy label window (NASA GFMS accessible period), with every feature present and a directly-observed positive flood label:

| Field | Value |
|---|---|
| Station / Date | SW90 (Bahadurabad, Jamuna) — 2021-05-24 |
| Local rainfall | 10.8 mm |
| Upstream basin rainfall | 5.7 mm |
| Soil moisture | 0.448 $m^3/m^3$ |
| River discharge | 23,119.6 $m^3/s$ |
| Flood within 72h? | Yes — directly observed by satellite (not inferred) |

### (b) Incomplete — a real, explicitly-flagged missing value

River discharge reanalysis only has real coverage from ~1997 onward. Rather than guessing a value for earlier dates, we leave it as a true missing value with an explicit flag our models are trained to handle natively — never a filled-in 0 or average, which would silently misrepresent the reading:

| Field | Value |
|---|---|
| Station / Date | SW267 (Sunamganj, Surma) — 1988-06-24 |
| Local rainfall | 270.9 mm (present) |
| Soil moisture | 0.475 $m^3/m^3$ (present) |
| River discharge | Missing (river_discharge_m3s_missing = True) |
| Flood within 72h? | Yes — but from a historical-archive match (event date range), not a same-day satellite detection, so we label its *regime* as "unobserved_positive" and weight it less heavily in training than case (a) |

### (c) Bad — data we identified as unreliable and excluded

Two real examples of data we caught and did not use, rather than trusting a source at face value:

| Case | What we found |
|---|---|
| A government report sentence | Parsing FFWC's 2012 Annual Flood Report, one sentence read: "attained the peak of 29.98m on 5th August which was 2cm below the DL(30.0m)" (Kaunia station, Teesta river). This sentence contains a date, so a naive parser would record it as a flood day — but the text says the water stayed *below* the danger level. We built an explicit check for this pattern and excluded 73 similar sentences (12% of our first parsing pass) before they could enter the training data. |

| | |
|---|---|
| A third-party public dataset | We found a GitHub-hosted dataset claiming to provide a daily "flood index" for 34 Bangladesh stations, 1980–2020. Before using it, we checked its actual value distribution and found the "index" simply followed the calendar monsoon season every year, with no year-to-year variation reflecting real flood events — it was a monsoon-calendar mask, not real flood data. We rejected it entirely rather than let the model learn to predict the calendar instead of real weather. |

# 3. Model 1 — Flood Risk Classification

## 3.1 Problem Definition

Our first model predicts, independently for each of 24, 48, and 72 hours ahead: will a flood (river level above the danger threshold) occur at this station within that window? This gives three separate binary probabilities per station per day rather than one combined answer, matching how real early-warning systems typically present multi-lead-time risk.

## 3.2 Features

- Local and upstream-basin rainfall (current value, 1/2/3/5-day lags, 7/14-day rolling sums, trend ratio)
- Local soil moisture (current value, lags, 30-day delta, and a Soil Wetness Index — an exponential recursive filter that better represents root-zone antecedent wetness than the raw surface reading)
- River discharge, both at the station itself and (where geographically meaningful) at an upstream in-network station and an upstream India-side reference point, each shifted by an estimated travel time
- Static terrain: elevation and Height Above Nearest Drainage (HAND), a peer-reviewed flood-susceptibility metric
- Station identity and river basin (as categorical features) and cyclical day-of-year (to capture monsoon seasonality)

## 3.3 Training Methodology

- **Model:** LightGBM gradient-boosted trees, one pooled model per horizon (all 30 stations trained together, station identity passed in as a feature) — chosen over training 30 separate models since several stations individually have too few labeled examples.
- **Split:** time-based, never random — training uses all data before 2024-01-01; testing uses only 2024 onward, and only rows where the label came from our most reliable source, to avoid an artificially inflated test score.
- **Sample weighting:** positive examples are weighted by class imbalance computed only from our most reliable label source (not the full sparse-label-inflated set), and positive-only-source labels are further down-weighted relative to primary-source labels, since they are coarser (dated to an event's date range, not verified day-by-day).
- **Decision threshold:** tuned to reach 85% recall on the test set for each horizon, rather than the default 0.5 — for a flood warning, missing a real flood is far costlier than a false alarm.

## 3.4 Results

| Horizon | ROC-AUC | PR-AUC | Precision @ 85% recall | Recall |
|---|---|---|---|---|
| 24h | 0.883 | 0.214 | 13.6% | 85.0% |
| 48h | 0.864 | 0.245 | 16.1% | 85.0% |
| 72h | 0.849 | 0.261 | 18.4% | 85.0% |

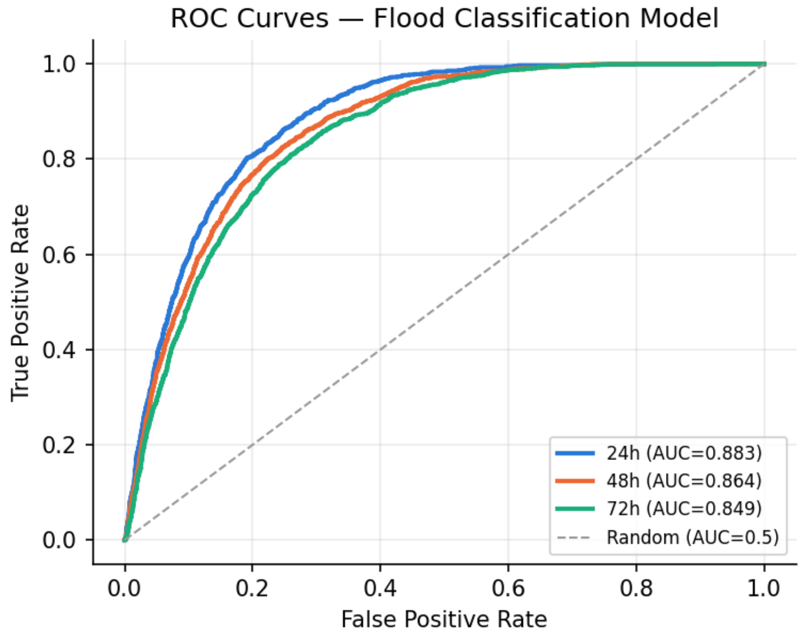


*Figure 3. ROC curves for all three horizons. All three sit well above the random-guess diagonal.*

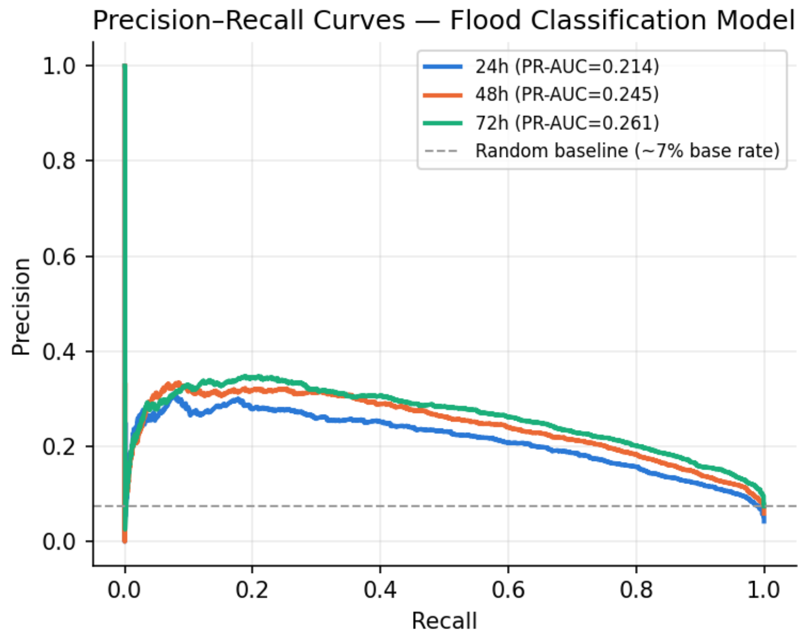


*Figure 4. Precision–recall curves. These sit far closer to the random baseline than the ROC curves — expected and disclosed below, not hidden.*

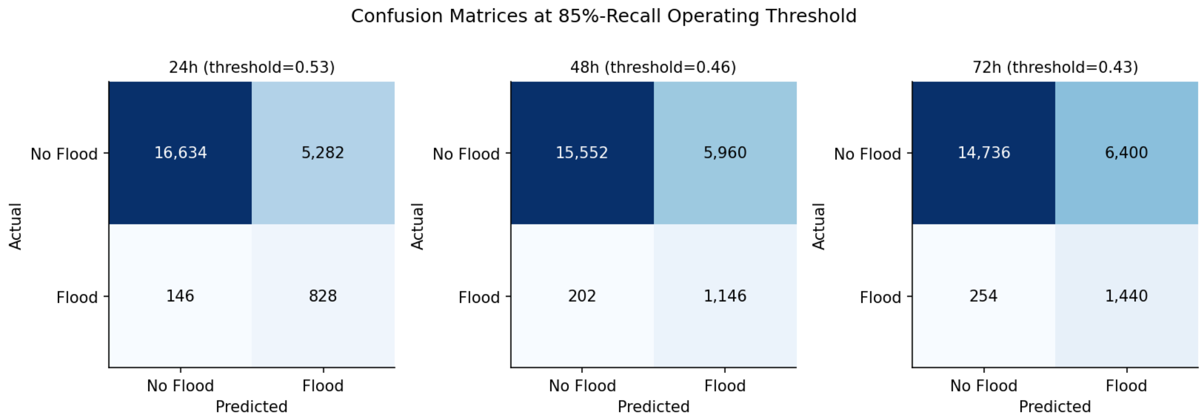


*Figure 5. Confusion matrices at each horizon's 85%-recall operating threshold.*

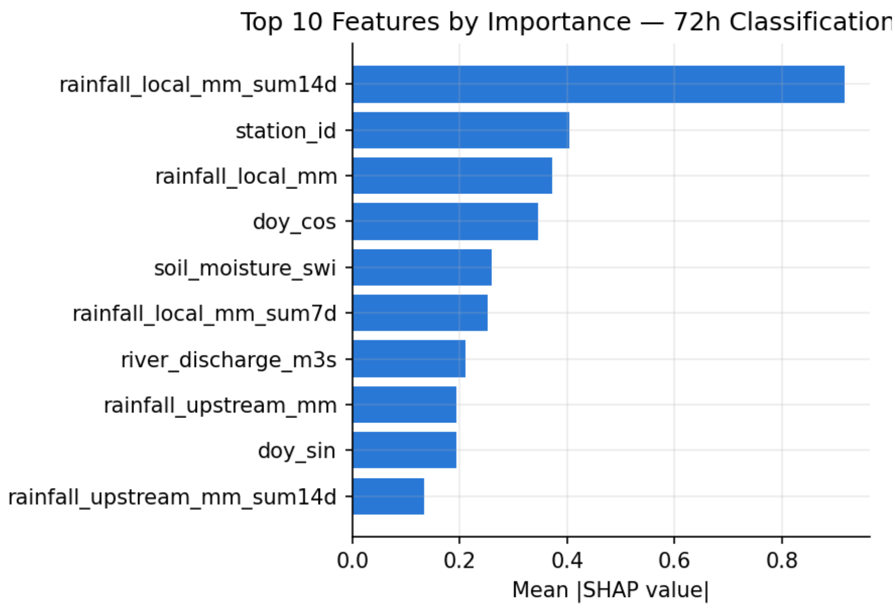


*Figure 6. Top 10 features by SHAP importance for the 72-hour model. 14-day cumulative local rainfall is the single strongest driver, followed by station identity and soil-moisture-based features.*

**Honest reading:** ROC-AUC around 0.85–0.88 shows the model genuinely separates flood-risk days from safe days. Precision at our chosen operating point is genuinely low (14–18%), meaning most "high risk" alerts will not be followed by an actual flood. This is a direct, structural consequence of two facts: positive examples are rare (roughly 4–7% of test days per horizon), and we deliberately tuned the threshold for high recall (catch real floods) rather than high precision (minimize false alarms) — the same tradeoff real weather warning systems make, since a missed flood is worse than an extra false alarm for a warning system's purpose.

# 4. Model 2 — River Discharge Forecasting (Pivot)

## 4.1 Motivation

Binary flood classification is fundamentally limited by how rare and sparse our positive labels are. River discharge, in contrast, is a dense, continuous measurement with real values on nearly every day since 1997 — no rare-event problem at all. As a second, independent model, we built a regressor that predicts a station's future river discharge (in cubic meters per second) instead of a binary flood flag, from which risk can be estimated more transparently (e.g. how far is the predicted flow from what is normal for this river at this time of year).

**This is a separate model, not a replacement** for the classifier in Section 3 — both exist independently, and we are seeking advice on which (or both, or how combined) is the better fit going forward.

## 4.2 Method

- Same feature set as the classification model (Section 3.2), reused directly rather than re-engineered.
- New targets: each station's own river discharge, 24/48/72 hours ahead.
- Target transform: log(1+discharge), because discharge spans roughly five orders of magnitude across our 30 stations (from ~2 $m^3/s$ on a small urban canal to ~39,000 $m^3/s$ on the combined Padma-Meghna) — without this transform, a plain squared-error model would be dominated entirely by the largest rivers.
- Same LightGBM framework and time-based train/test split as the classification model, for a fair, comparable evaluation setup.
- Every result is compared against a naive "tomorrow will be the same as today" persistence baseline — discharge is highly autocorrelated day-to-day, so a model must be measured against that naive guess to show it has learned anything beyond "assume no change."

## 4.3 Results

| Horizon | Model MAE ($m^3/s$) | Persistence MAE ($m^3/s$) | Improvement | $R^2$ |
|---|---|---|---|---|
| 24h | 321 | 361 | +11.1% | 0.996 |
| 48h | 534 | 674 | +20.8% | 0.989 |
| 72h | 672 | 937 | +28.2% | 0.983 |

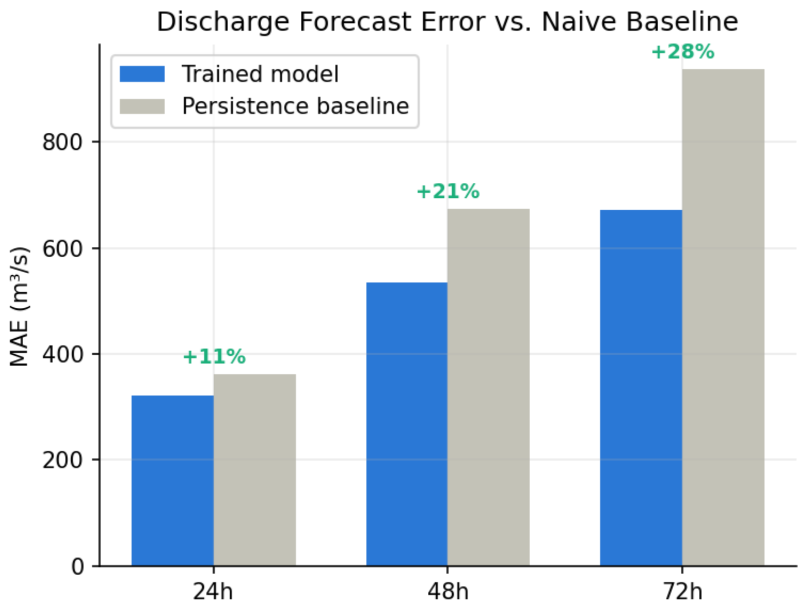


*Figure 7. Forecast error (MAE) vs. the naive persistence baseline. The model's advantage over persistence grows with horizon — 11% at 24h, up to 28% at 72h — the expected shape for a model with real skill, since persistence gets worse the further ahead it is asked to guess.*

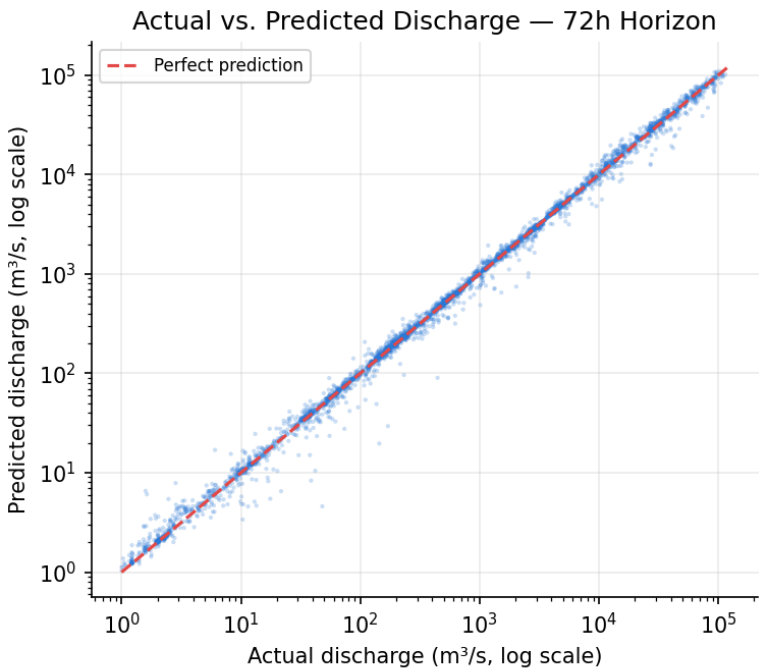


*Figure 8. Actual vs. predicted discharge, 72-hour horizon (log-log scale, sampled test points). IMPORTANT CAVEAT: this looks almost perfect, but that is significantly a visual effect of the log-log scale compressing five orders of magnitude of real variation — a model that only gets each river's rough scale right already looks this good on this kind of plot. Figure 7's persistence comparison, not this scatter plot or the $R^2$ figures above, is the fairer measure of real added skill.*

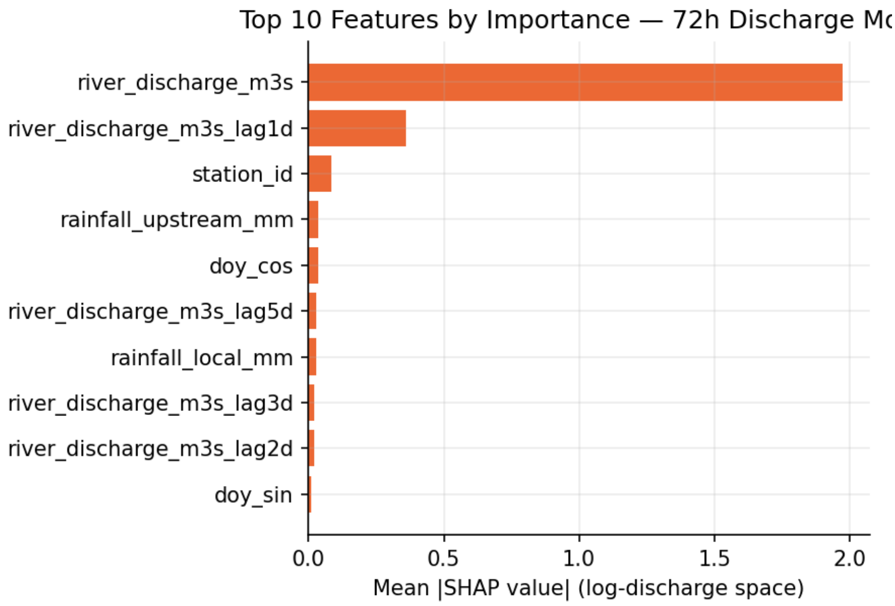


*Figure 9. Top 10 features for the 72-hour discharge model. Today's own discharge dominates (expected — real physical persistence, not a data leak, since it is genuinely known at prediction time), followed by yesterday's discharge, station identity, then rainfall.*

**Honest reading:** $R^2$ of 0.98–0.996 looks excellent but is substantially inflated by the huge scale difference between stations — we do not treat it as the headline number. The metric we trust most is the persistence comparison in Figure 7: a real, growing advantage over the naive baseline, which is the right shape for genuine forecasting skill rather than an artifact of scale.

## 5. Live Serving & Validation

Beyond training, we built a live prediction pipeline: given a location, it fetches current rainfall, soil moisture, and river discharge from free live APIs, assembles the same features used in training, and returns a prediction from the classification model for all three horizons. We hardened this pipeline against real failure modes (malformed upstream API responses, out-of-region coordinates, unexpected internal errors) by deliberately testing each one, not just reasoning about it.

We also validated the live model against FFWC's own public real-time flood map by hand: at the time checked, our model agreed directionally with every station FFWC currently listed as elevated risk ("Warning" or "Flood"), but also frequently predicted "high" risk for several stations FFWC listed as "Normal." We believe this is consistent with the classification model's disclosed low precision (Section 3.4) rather than a new problem, but flag it honestly as unconfirmed at this point — see the open questions on the final page.

# 6. Limitations, Problems, and Questions for Our Advisor

We are documenting these honestly because we would like guidance on how to prioritize them, not because we believe the work so far is unsound.

## 6.1 Known Limitations

- **Low precision at our operating threshold (14–18%).** Most "high risk" alerts from the classification model will not be followed by an actual flood. This is a structural consequence of how rare flood events are in our label data, combined with deliberately prioritizing recall — not a bug, but a real cost of the design choice.
- **No live official water-level feed yet.** FFWC's real-time API requires manual registration (an email request is pending); until then, our live predictions rely on weather/discharge proxies only, not the official gauge readings themselves.
- **No forecast-rainfall feature.** Our models were trained purely on already-observed (backward-looking) data — they do not currently use tomorrow's rain forecast as an input, even though it is available live. Adding it would require a retrain and a source of historical forecast data for training, which is harder to obtain than historical observations.
- **The discharge regression model's real-world usefulness as a flood indicator is unproven.** We do not have each station's official "danger level" in the same discharge units (only in water-level meters, from FFWC), so we cannot yet directly say "this predicted discharge means flooding" — only that discharge is trending unusually high or low.
- **Coarse geographic approximations.** Our 30 station coordinates are real-town/confluence approximations, not exact FFWC gauge locations, and our discharge source (GloFAS, a ~0.05° global reanalysis grid) occasionally lands on the wrong channel near confluences — we manually corrected this where verifiable, but two stations remain flagged as uncertain.
- **Sparse label years.** 2017–2020 in particular has much thinner labeled coverage than other years, despite deliberate effort to patch it from four additional sources.

## 6.2 Questions We Would Like Advice On

### Q1. Precision vs. recall tradeoff:

Is prioritizing recall (fewer missed floods, more false alarms) the right choice for this system, or should we retune toward higher precision given this will run alongside an automated IoT system rather than only inform a human? We are concerned about false-alarm fatigue if this is wired to trigger automated actions.

### Q2. Which model should be the primary deliverable?

Should we present the classification model, the discharge regression model, or both as co-equal components? The regression model has a more defensible evaluation methodology (dense data, honest baseline comparison) but does not directly answer "will it flood."

### Q3. IoT integration design:

We are considering two designs: (a) an advisory-only dashboard showing the model's output alongside sensor readings, with no automated action, or (b) a combined-condition trigger where the motors act pre-emptively only when both the regional model AND our own local ultrasonic sensor agree. Which is more appropriate for a final-year project given the model's known precision limits?

### Q4. Is the live/FFWC disagreement acceptable to leave as-is?

Our live validation (Section 5) found the model frequently predicts elevated risk for stations FFWC currently calls "Normal." We believe this reflects the known precision tradeoff rather than a hidden bug, but we have not yet built a way to measure this systematically over time. Should we prioritize building that measurement before anything else?

### Q5. Scope for remaining time:

Given our remaining timeline, which of the following is the best use of time: pursuing live FFWC data access, adding the forecast-rainfall feature, building the IoT integration layer, or further validating the discharge model as a flood indicator?